MACHINE LEARNING PROJECT

Spotify Genre Classification: Predicting Music Genre from Audio Features

A supervised multiclass classification approach using scikit-learn

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https://github.com/Claffyhill/spotify-genre-ml

Problem Statement

Multiclass supervised classification across 114 balanced genre classes

Objective

Given the numerical audio features of a track, predict its genre (track_genre). This constitutes a multiclass supervised classification problem with 114 balanced target classes.

- Input (X): 14 numerical audio features — popularity, duration, danceability, energy, loudness, and others
- Output (Y): one of 114 mutually exclusive genre labels
- The class set is balanced (~1,000 tracks per genre), ensuring no majority-class bias
- Predicting among 114 genres is substantially harder than binary classification: random accuracy is below 1%

114

GENRE CLASSES

Compared to a typical binary classifier (2 classes), a 114-class problem imposes considerably greater demands on model capacity and evaluation. Chance-level performance corresponds to approximately 0.88% accuracy.

Dataset Description

Spotify Tracks Genre Dataset (Kaggle)

114,000	TRACKS
21	COLUMNS
114	BALANCED GENRES
No	MISSING VALUES
No	DUPLICATED ROWS

14 numeric input features (X)

popularity	duration_ms	danceability
energy	key	loudness
mode	speechiness	acousticness
instrumentalness	liveness	valence
tempo	time_signature	

Excluded: track ID, artist name, album name, track name — text metadata was not used as model input

Exploratory Data Analysis (EDA)

Understanding the data before modeling

Integrity checks

Verified shape, data types, missing values, and duplicates — the dataset required no cleaning.

Class distribution

Confirmed 114 unique genres with a balanced distribution (~1,000 tracks per class).

Feature subset

Identified the 14 numeric variables to be used as input, excluding IDs and text fields.

Feature separability

Examined feature distributions across top genres to assess inter-class separability.

Key finding

Some genres exhibit visually distinguishable feature patterns — for instance, acoustic tracks tend toward high acousticness and low energy, while electronic genres show the inverse profile. However, a substantial number of genre pairs overlap considerably in feature space, suggesting that numeric audio features alone may be insufficient to achieve high accuracy across all 114 classes. This observation motivated the comparison of both linear and no n-linear classifiers.

Preprocessing Pipeline

From 114,000 raw tracks to a model-ready dataset

1

Stratified subsampling

A stratified sample of 20,000 tracks was drawn from the full dataset, preserving the original genre distribution across all 114 classes.

→

2

Train / test split

The sample was partitioned into 75% training (15,000 tracks) and 25% test (5,000 tracks), stratified by genre label.

→

3

Feature scaling (StandardScaler)

Applied to Logistic Regression and KNN. Logistic Regression relies on gradient-based optimization sensitive to feature magnitude; KNN uses Euclidean distance, which is distorted by differing scales. Random Forest is scale-invariant and was trained on unscaled data.

All steps used a fixed random_state to ensure reproducibility of train/test splits and model training.

Feature Selection

What features enter the models, and why

Selected features

- All 14 numerical audio features were retained as model input.
- Text metadata (artist, album, track name) and identifier columns (track ID) were excluded, as they do not represent audio signal and would not generalise to unseen tracks.
- No dimensionality reduction technique was applied.
- Given that the dataset contains only 14 numerical variables, the feature space is already low-dimensional and formal feature selection was not necessary for this iteration of the project.

Rationale

With only 14 input variables, dimensionality is not a concern and explicit feature selection was unnecessary.

All 14 features used
No selection applied

Model Selection

Four classifiers across three learning paradigms — selection rationale

Dummy Classifier

BASELINE

Why selected:

Provides a performance lower bound. Predicts the most frequent class regardless of input. Any useful model must substantially outperform this baseline.

Logistic Regression

LINEAR

Why selected:

Simplest trainable linear classifier. Establishes whether a linear decision boundary is sufficient to separate genres in the 14-dimensional feature space.

K-Nearest Neighbors (k = 5)

DISTANCE-BASED

Why selected:

Instance-based, non-parametric classifier that captures local similarity. Included to assess whether genre proximity in feature space is a reliable signal.

Random Forest (100 trees)

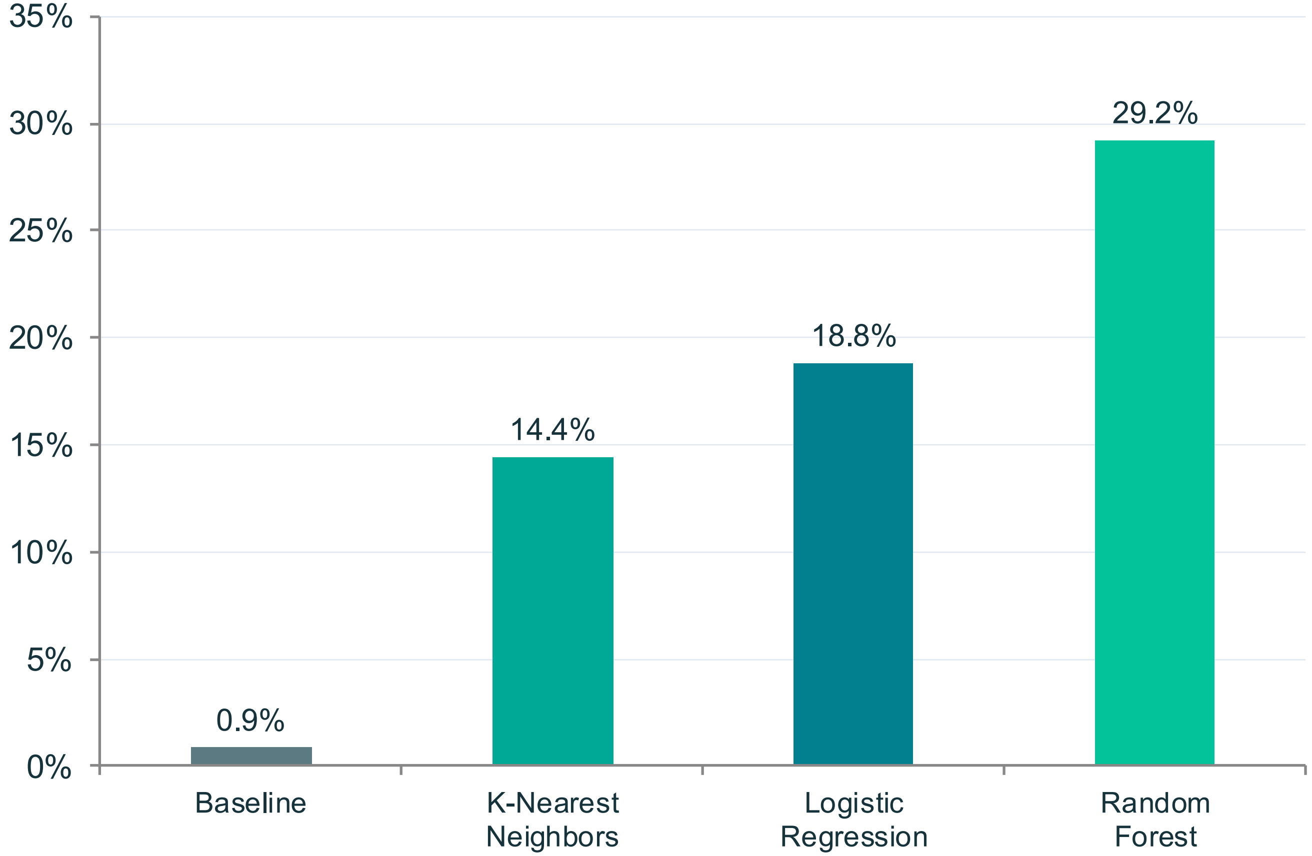
ENSEMBLE

Why selected:

Ensemble model capable of learning non-linear relationships between audio features. Expected to outperform linear models given the complex, overlapping class structure observed during EDA.

Training & Evaluation

Identical training and test conditions across all models



Evaluation setup

- Training set: 15,000 tracks
- Test set: 5,000 tracks
- Primary metric: accuracy
- Secondary metrics: precision, recall, F1-score per class
- Confusion matrix on the top 10 most frequent genres

Random Forest

Random Forest achieved the highest accuracy (29.18%), outperforming the other evaluated classifiers.

Results: Metrics & Interpretation

Per-class performance and contextual interpretation of accuracy

Genre	Precision	Recall	F1-score
comedy	0.89	0.75	0.81
grindcore	0.75	0.86	0.80
honky-tonk	0.78	0.70	0.74
emo	0.00	0.00	0.00
electronic	0.06	0.02	0.03

Macro avg: Precision 0.28 · Recall 0.29 · F1 0.28 | Weighted avg ≈ 0.28

Overall, the results indicate that most prediction errors occur between acoustically similar genres rather than unrelated ones.

Interpretation

- Although the overall accuracy is 29.18%, the task involves distinguishing among 114 balanced classes, making it considerably more challenging than conventional binary classification problems.
- Best-classified genres — comedy (F1 = 0.81), grindcore (F1 = 0.80), honky-tonk (F1 = 0.74) — exhibit acoustically distinctive and non-overlapping feature profiles.
- Most-confounded genres (emo, electronic, spanish/malay) share feature distributions with closely related styles, resulting in near-zero F1 scores.
- The confusion matrix reveals that misclassifications are concentrated within genre families (e.g., house / deep-house / techno), not randomly distributed — confirming that errors reflect genuine acoustic similarity.

Conclusions & Future Work

CONCLUSIONS

1. Numerical audio features encode a learnable signal for genre prediction — all trained classifiers substantially outperform the random baseline.
2. Random Forest was the most effective model, benefiting from its ability to capture non-linear feature interactions without requiring feature scaling.
3. Classification errors are concentrated within acoustically similar genre families, indicating a natural upper bound for approaches that rely exclusively on numeric audio features.

FUTURE WORK

Hyperparameter tuning

Systematic search over Random Forest parameters (n_estimators, max_depth, min_samples_leaf)

Feature engineering

Incorporate additional metadata or engineered audio features.

Additional ensemble methods

Evaluate Gradient Boosting variants on the same train/test partition

Neural network classifiers

Apply shallow feed-forward networks to explore non-linear capacity beyond tree ensembles

Thank you — questions welcome